ICT Unit C: Computer and its Applications

Ch16 The Networking and Internet Basics

Concept.1 Computer network

- **Computer network**: System of interconnected Computers + Peripherals + Communication devices
  ⇒ Exchange Information + Shared resources

➢ LANs vs WANs

- LANs (Local Area Network) ⇔ Computers , meaning a relatively limited area.

Characteristic	Peer-to-peer network	Client / server network
Central server connection	✕	✓
Location of shared resource	Peer workstations	Central server
Number of workstations	Small	Large
Network loading limit	Lower	High
Network security control	Poor	Good
Cost	Low	High

- WANs (Wide Area Network) ⇔ LANs , meaning a wide geographical area.
- E.g. The **internet** is the largest **WAN** in the communication world

Concept.2 Network services

	Service	Description	Remark
Communication	1. Internal communication	Communication between computers	E.g. e-mail, instant message, file transfer
	2. Network conferencing (Network communication)	2-way audio & video signals	Video signals must be compressed ✓ Instant and cost saving
Sharing	3. Hardware sharing	LAN: sharing hard disks, printers and scanners WAN: hard disk storage space	✓ Better monitoring and utilization ✓ Saving costs
	4. Sharing of network licensed software	Among computers simultaneously	✓ Lower cost ✓ Saving time for installation
	5. Internet access	Access the internet via a computer network	✓ Lower cost ✓ Better monitoring + network protection
Data	6. Centralized data management	Centralize data in a network server **Security control**: Firewalls / **VPN** (Virtual Private Network) / Dedicated leased lines / Virtual Connection Linkage (Tunnel)	✓ Keep information updated ✓ Better security control ✓ Increasing productivity and efficiency
	7. Information backup	Backup information on individual computers to a server via a network	For rapid system recovery

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Concept.3 Network connection

1. **WLAN**: Connected to a wireless LAN by means of an access point
2. **Access point** allows a wireless device to: Communicate to other wireless devices (Radio frequencies)
   Connect wired network (Radio ⇔ Digital signal)
3. **Wireless NIC** (Network Interface Card): Connect a device to a WLAN

Network types	Mechanism
1. Satellite broadband	Establish a microwave connection to the internet for remote areas (✕ Wired broadband) (✕ Radio ∵ Radio will be reflected by the ionosphere)
2. Microwave broadband	**Access point** (On top of building) + **Subscriber units** Subscriber unit must have a line-of-sight to the access point
3. Wireless Fidelity (Wi-Fi)	Conversion between digital data and radio signals **DTS**: 11Mbps (802.11b) ~ 600Mbps (802.11n) **Distance**: Indoor: 38m (802.11b) ~ Outdoor: 250m (802.11n)

Features

Advantage	Disadvantage
1. Network set-up cost ↓ 2. Flexible network structure 3. Can extend network to an outdoor area	1. Unstable data signal 2. Risk of unauthorized intrusions ↑ 3. DTS (Data Transmission Speed) < that of a wired network 4. Limited signal coverage 5. Cost of setting up a large wireless network ↑

Comparisons of communication links

	Twisted-pair cable	Coaxial cable	Optical fibre cable	Wireless (IEEE 802.11 standard)
DTS	10Mbps, 100Mbps	~ 299Mbps	100Mbps to 10 Gbps	11Mbps ~ 600Mbps
Transmission signal	Electrical	Electrical	Optical	Radio waves
Cost	Cheap	Expansive	Most expensive	∝ Network size

[1] WAP: Wireless Application Protocol

[2] GPRS: General Packet Radio Service

[3] ISDN: Integrated Service Digital Network

[4] ADSL (A)symmetric Digital Subscriber Line (Upstream 640kbps < Downstream 9Mbps)
SDSL (Upstream = Downstream 24Mbps)

[5] Need to share cable network in the same location (30Mbps)

[6] Dial-up connection: 56kbps(Vocal signal) | DSL modem: 8Mbps (Downstream) + 1Mbps (Upstream) (Data transmission)

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Methods of internet access

	Broadband	Wi-Fi hot spot	WAP[1]	GPRS[2]	ISDN[3]	Dial-up connection[6]
DTS	1.5Mbps ~ 1000Mbps	11Mbps ~ 600Mbps	10Kbps ~ 153Kbps	Max. 115.2Kbps	Max. 128kbps (digital)	Max. 56kbps (Analogue)
Cost	High	Very high			Average	Low
Availability (Area covered by)	ADSL[4]: Telephone network Cable[5]: Cable TV network	Wi-Fi access points	Mobile phone network		Telephone network	
Security	High	Low	Average		Very high	High (∵ Analogue)
Application	• Households • Schools • Large organization	• Travellers • PDA user	• Smart phone • PDA	• Smart phone • PDA • notebook computer	• Schools • Company • Large organization	• Remote households • Travellers
Popularity	Majority	Moderate	Minority			

Concept.4 Network hardware

➢ Expansion card & connectors

- **Network Interface Card (NIC)**: Expansion board → Connect a computer to a network

  Each NIC has unique **media access control (MAC) address**: Identify a computer on the network

  1. Connector — BNC and RJ-45
  2. Expansion bus — ISA, PCI, PCMCIA and USB
  3. Bandwidth — 10Mbps, 100Mbps, 1000Mbps (1Mb = 1000Kb ≠ 1024Kb)

➢ Connection devices

Device	Function		Application
1. Switch	Central connection point	High data transmission speed (10Mbps ~ 1Gbps)	Computers ⇔ Servers
2. Hub		Low data transmission speed	
3. Bridge	Connects networks	LAN A1 ⇔ LAN A2	⇔ LANs (Same type)
4. Router (Translation)		Network A ⇔ Network B	LANs ⇔ Internet Home-use router: Switching router
5. Modem (Conversion)	(Type.1) Dial-up modem	Digital signals ⇔ Analogue Telephone signals	Dialup connection to a network
MO: Modulation	(Type.2) Cable modem	Digital signals ⇔ Analogue cable TV signals	Broadband service via TV cables
DEM: Demodulation	(Type.3) Digital subscriber line (DSL) modem	1. Filtering DSL signals 2. DSL signals ⇔ Signals that a computer can process	Broadband service via telephone cables

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6. Repeater	1. **Amplifying** received signals **2. Removing unwanted noises** 3. Sending the amplified signals to the connected device	Connection of remote deivces

➢ Transmission medium – Cables

Types	Remarks		Applications
1. Twisted-pair cables (Twisting → EM interference ↑	Shielded TP (STP)	With metal wrapper	Cat 1 ~ Cat 6 (Maximum data transmission speed (DTS): 1Gbps) (Distance: 100m)
	Unshielded TP (UTP)		
2. Coaxial cables	1. Noise ↓ 2. DTS ↑ 3. Distance ↑		VCR, cable TV connection

4. Optical fibres	Advantage	Disadvantage
	1. Signal-carrying capacity ↑ **2. Data transmission speed ↑** **3. Security for transmission ↑** **4. Thinner + Lighter + Durable**	1. Cost ↑ 2. Difficult to install + change 3. Require a device for signal conversion

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Ch17 Communication Software and Protocols

➢ Communication software and protocols

- **Computer network** → Information sharing + Resources sharing + Communication (Users ⇔ Users)
  - ✓ Data transmission speed ↑
  - ✓ Cost of transmission ↓
  - ✓Support multimedia message
  - ✓ **Reusability of transmitted information** ↑
- **Communication software** → Transmit digital messages through a computer network (Users ⇔ Users) → Utilizes communication protocol → Work with communication devices
- Communication protocol: A set of rules ∈ data **formatting / signaling** → understood by all the communication devices

Communication protocol	Application
Transmission Control Protocol (TCP) / Internet Protocol (IP)	Network communications over the Internet and Intranet
Simple Mail Transfer Protocol (SMTP)	Transferring e-mail messages
Wireless Application Protocol (WAP)	Wireless communications for handheld devices
Voice over Internet Protocol (VoIP)	Voice data transmission by Internet phone communication software
Telnet protocol, Secure Shell (SSH)	Establish connection for remote logon (Computers ⇔ Servers)

HTTP (HyperText Transfer Protocol): Port 80,

FTP (File Transfer Protocol): Port 20,

SMTP: Port 25, **Telnet**: (Data Port) Port 23 (1/number ∝ Security level)

[1]Transportation information: Protocol

[2]3 Network address information: IP

[3]Data Link Header: Physical address + Method of error detection& correction

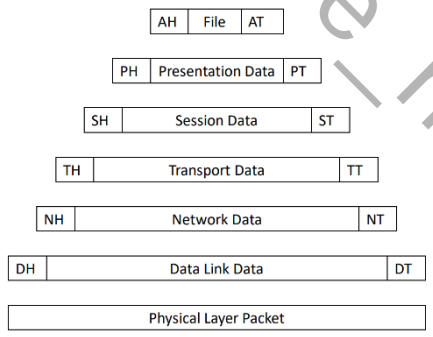
[4]Data Link Trailer: String of bits ◊ Indicates the end of the packet

➢ OSI Model – Open System Interconnection Model

Layer name	Key responsibilities (Sender's aspect)	Data type handled	Protocols and technology
1. Application	Interface provision → Set up communications ⇔ application software	User data	DNS; FTP; SMTP; POP3; IMAP; HTTP; Telet
2. Presentation	Data format conversion → Data: can be transmitted + compatible (receiver's system)	Encoded user data	SSL (Secure Sockets Layer)
3. Session	Set up and maintain the communication session (during data transmission)	Sessions	H.245
4. Transport	Data + Transport Header(TH)[1] → Datagram	Datagrams / Segments	TCP
5. Network (add IP address)	Datagram + Network Header(NH)[2] → Packet	Datagrams / Packets	IP
6. Data link (add physical address)	Determine network addressing Error detection + correction Header(DH)[3] + Trailer(DT)[4] + Packet → Frame	Frames	IEEE 802.11 (WLAN, Wi-Fi)
7. Physical	Physical transmission of Frame (through LAN)	Bits	

PDU = Protocol Data Unit (a packet)

Application Later: PDU
Presentation Layer: PDU
Session Layer: PDU
Transport Segment
Network Datagram
Data Link Packet
Physical Bits



➢ Network and Transport Layer – TCP/IP

- **Communication protocol suite**: TCP/IP → Construct the Internet + computer network

  ⇒ Send data over a **Packet Switching Network**

TCP layer	Receiver's aspect	Resembles the received data packet's into 1 data file
	Sender's aspect	1. **Divide** data → Packets 2. Provide a **unique number** to each packet 3. Forward packets to the IP layer
IP layer		1. Each packet **add** destination IP address **2. Packet fragmentation** 3. Select **the shortest path** to the destination

- **IP address**: Identify computers on a network
  - IP address = (Class number +) Network address[1] + Host address[2]

[1] Identify a particular **network** [2]Identify a particular **device** in the network

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IPv4 – Total number: $(2^8)^4$ e.g. 127.0.0.1 = 01111111_00000000_00000000_00000001 (Class A)

Class D and E are not used in normal operation

Class	Class number (✕ leftmost '0')	Partition of IP address				Number of hosts
A	0	Position	0	1~7	8~31	2^{24}- 2
		Part	Class number	Network address	Host address	
B	10	Position	0~1	2~15	16~31	2^{16}- 2
		Part	Class number	Network address	Host address	
C	110	Position	0~2	3~23	24~31	2^8 - 2
		Part	Class number	Network address	Host address	

IPV6 – Total number: $(2^{16})^8$ e.g. 1010:0:0:8:800:2000:4:300

➢ Application layer – DNS (Domain Name Server)

- **Domain name**: Fully Qualified Domain Name (FQDN)
- **DNS**: Translates domain names → IP address
- **URL** (Uniform Resource Locator): Particular web page / file (on a website)

2nd level domain Top level domain (TLD)

(1st level domain name)

Generic name (e.g. com, org, net)

+ Country code (e.g. cn, tw, hk)

http	www	google	com	google-earth	Readme.html
Protocol	(www. Host name) Domain name			Folder name	Document name

- E.g. for ftp (file transfer protocol): ftp://index.html <alanshum>@<sample domain name>/…
- **Network solution** (inc.): Store the domain name
- HKIRC: Manage the registry of domain name with country code = 'hk'

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Ch18 Services and Applications

Concept.1 Information digging

➢ Obtain – Information search

- **Search engine**: Coordinated a set of programs (Spider / Bot + Spider algorithm → Finds websites + web pages on WWW

  ← Match the keywords (Parameter for search engine → Perform searching) with the content of web pages

Search engine types	Example
1. General search engine	Google.com , Yahoo
2. General search engine using natural language	Ask.com
3. Specialised search engine	Bloomberg.com
4. Meta-search engine (powered by other engine)	WebCrawler

Boolean search: AND, OR, NOT; "…": search a whole sentence; …*: wildcard

➢ Analysis – Information sources

Completeness	Accuracy	Authority	Objectivity	Timeliness	Legality
Comprehensive	Correct	Credible	Objective	Updated	Legal to use + store

Concept.2 Internet services

- **Web browsing** is referred to as navigating, reading and interacting with web sites on the WWW.

Services	Explanations / Functions	
HTTP	Used by a program (in web server) → Exchange data on WWW (≠WAN)	
Graphics	Bitmap	JPG , GIF , PNG , BMP (Uncompressed)
	Vector graphics	WMF , SWF , EPS
Audio	Compressed	WMA , MP3 , AAC
	Video streaming	WMA
Video	Video streaming	WMV , RM , MOV , MP4 (✕ .avi)
Plug-in (small program)	Integrated into a web browser → Provide additional functions • Standalone web browser size ↓	
FTP	Transfer files (Computers ⇔ Computers) by **user authentication** / **anonymous FTP** Send files: **ASCII** (Send plaint text) / **Binary file** (Send but-by-bit)	
Email	Transmission of digital documents (Computers ⇔ Computers)	
	Protocol	SMTP , POP , IMAP
Newsgroup	Read + Post text messages (Text + Computer graphics + Hyperlinks)	
Discussion forum	≈ Newsgroup + Post messages (Text + Computer graphics + Hyperlinks)	
Online chat	1-to-1 / 1-to-many discussions by voice communication / messages (Text + Graphics)	
Remote logon	Establish a connection (Computers ⇔ Server) via the internet / public communication media	
	Protocol	Telnet , Secure Shell (SSH) (More secured ∵Encryption)

Abbreviation

AAC	Advanced Audio Coding	**PNG**	Portable Network Graphics
EPS	Encapsulated Postscript	**SWF**	Shockwave Flash
FTP	File Transfer Protocol	**WAV**	Waveform Audio Format
GIF	Graphics Interchange Format	**WMA**	Windows Media Audio
JPEG	Joint Photographer Expert Group	**WMF**	Windows Metafile
MP3	MPEG-1 Audio Layer 3	**WMV**	Window Media Video

Concept.3 Internet Technologies & Application

➢ Internet - Technologies

- **Streaming technology**: ✓ Download a portion of audio / video file + play simultaneously
- Applications: Webcasting sites, voice mail, video conferencing and some peer-to-peer phone services

Types	Explanations
1. Webcasting	Broadcasting of multimedia content using streaming technology
2. Voice mail services	Send voice messages to offline users
3. Video conferencing	Real-time video display (Interactive communication)

[1]ETF(Electronic Funds Transfer): System of transferring money (Bank AC 1 ⇔ Bank AC2) through computer network

[2]EDI(Electronic Data Interchange): Set of data transmission standard → Control electronic documents

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➢ Internet – Applications

Applications	Descriptions		Example
1. E-commerce	Business-to-Consumer (B2C)	Business activities using EFT[1] (Using telephone line + EDI[2] → Accuracy ↑ → Administrative cost ↓	Taobao, Yahoo
	Business-to-Business (B2B)		
	Consumer-to-Consumer (C2C)		
2. E-government	Services quality ↑ + Delivery efficiency + Effectiveness ↑		Gov.hk
3. E-learning	• Flexibility ↑ + Cost ↓ • Facilitate learning through computer + communication technologies • ✓ Latest knowledge + Skills		Web-based learning / training
4. Online entertainment	• Flexibility ↑ + Cost ↓ • Provide pleasure / relaxation • Censored contents		Online computer games / Music

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Ch19 Introduction to HTML

HTML Hypertext Mark-up Language:

- **Marked-up-language**: Define properties of elements on a webpage
  ⇒ OS independent → Cross-platform → Portable
  ⇒ ✕ Programming language
- **HTML document**: Plain text with HTML tags (✓ ASCII codes only)
  - **HTML tag** (Case-insensitive: Define appearance + Properties of different elements on a web page
    ✓ Enclosed by '<>', angle-bracket
  - **URL** (Uniform Resources Locator): Element of HTML document
  - **Hypertext**: A text that ca link to another document

HTTP:

- Hypertext Transfer Protocol (Application protocol)
  ⇒ Transfer hypertext document (HTTP server → Client)
- **HTTP request / Response message**

HTTP GET	Request resources on a web server
HTTP POST	Submit data to the server to be processed
HTTP HEAD	Request the header of the web page

Webpages:

- Electronic document written in hypertext format → Interpreted + Display by a web browser (HTTP client)
  1. ✓ Arithmetic calculation 2. Pure HTML code: ✕ conditional statement (e.g. IF)

➢ Basic HTML Tags

Attributes in <BODY> - Define elements of the whole webpage

text = "abc"	If abc = #00FF00"	Set the font colour
	If abc = Times New Rome	Set the font face
background = 'xxx.jpg'	Set the background image → xxx.jpg	
bgcolor = "orange"	Set the background colour → orange	

Other Tags can also be embedded in the <BODY …> tag

<font size = 7>	Set the font size
<embed src = "xxx1.mp3></embed>	Set the background music to xxx1.mp3 (played automatically)

➢ Document layout

Tag	Representation	
<H1>…<H1> to <H6>…<H6>	Section header	<H1> is greater than <H6>
** **	Line break	Break a new line
<CENTER>…</CENTER>	Centre alignment	
<FRAME>…</FRAME>	Specify the web page to be initially loaded into that frame	

Attributes in <FRAME>

name = "xxx"	**Name of the frame**	
scrolling = "xxx"	If scrolling is allowed	"auto" / "yes" / "no"
Frameborder = "x"	If the frame has a border	"1" / "0"
marginwidth = "x"	Width of the frame	
marginheight = "x"	Height of the frame	
src = "xxx"	The link of the web page	
target = "xxx"	Where the clicked link should go (e.g. "_parent", "_self", "_blank", "_top"	

➢ Text format

Tag	Representation	Tag	Representation
<B>…</B>	Bold	**[…]**	Superscript (e.g. ^{1})
<I>…</I>	Italic	**_…**	Subscript (e.g. $_{1}$)
<U>…</U>	Underline	**<FONT>…</FONT>**	Font (face, size, colour)

➢ Table

Tag	Representation
<TABLE>…</TABLE>	Table
<TR>…</TR>	Table row
<TH>…</TH>	Table header

➢ List

Tag	Representation
<OL>…</OL>	Ordered list (e.g. 1. 2. 3.)
<UL>…</UL>	Unordered list (e.g. •)

➢ Image

Tag	Representation	
<IMG>…</IMG>	Inline image	</IMG> is not needed
<A>…</A>	Anchor	

Attributes in <IMG> : src="xxx.jpg" is the image source

- If the image size < screen resolution ⇒ the images will be duplicated to fill up the screen

Attributes in <A>: href = "<html doc>" is the hypertext reference ; name = "xxx" is the name of the anchor

- Link the anchor of href by using the name of another anchor: <A href = "#example1">

1. We need a '#', hash symbol
2. Two anchors should be on the same web page

➢ Colour representation

- Colour name: Common colour names (e.g. orange, white, blue)
- Six-digit hexadecimal colour code #xxxxxx (e.g. #FF00FF = Red (255) + Blue (255)

Ch20 Elementary Web Authoring

➢ Web authoring (process)

- Establishing a **web site** (collection of related web pages) → Organized + Linked together
  - **Major parts in web authoring**: Designing **structure (内)** + **layout(外)** of web site
- **Copyright materials**: Requires written permission from the owners
- **JavaScript**: Scripting language (✕ Programming language) (✕ Object-oriented language)
  ∵ Usually executed on coordination with applications

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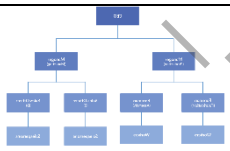
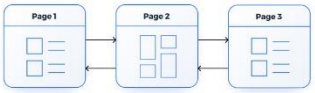
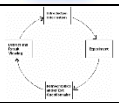
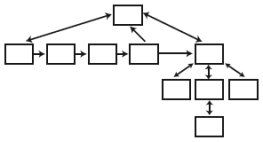
➢ Planning & Organizing a website

Phases (Website)	Workflows	Details
1. Planning	• Identify an intended **audience** • Clear **objectives** for the web site • Design the **information structure** of the web site (內) • Design the **Content + Layout** of each web page (外) • Prepare **materials** for the web pages	
2. Building (*IDE = integrated development environment)	• Use an IDE authoring tool • Appropriate multimedia elements • Categorize web contents	• Avoid **lengthy** page • **Uniform** design • **Hyperlinks** for web site navigation • Load within a **reasonable time** • **Updated**
3. Publishing & Testing	Web site publishing	Upload all web pages + Required files → Web server
	Web site testing	• Identify broken & missing hyperlinks • Scan for outdated files • Test loading time for web pages + linked files

➢ Relative path & Absolute path

Relative path **<A href="…/floder/file.html>**	• '/': For sub-folder / file in the same folder • '…/': For folder from the foot
Absolute path **<A href = 'https://www.abc.com>**	• Risk of influence caused by domain name, folder name or …

➢ Structure – Information organization

1. Hierarchical structure (Most common)		Homepage = Top-level web page of website = index.html = Parent page of all web pages Low-level web pages = Child page
2. Linear structure (Most rare) (Best for self-learning web site)		Organize information in sequential order: Simple + Clear
3. Cyclic structure		With sequence of web pages in iteration order
4. Composite structure		Hierarchical structure + Cyclic structure

➢ Basic operation of Adobe Dreamwaver CS3 in website development

• Adobe Dreamweaver: Web authoring tool → Design web pages + Managing web sites visually
  ✓ Tools → Automating web page design + Website maintenance
• Processes of building a website with Adobe Dreamweaver CS3:
  1. Defining – Specify name & local root folder for a new web site
  2. Managing – Create web pages + Construct website's structure; Manage files in both local machine & remote site; Keep track of any updates / broken links among files
  3. Designing – Tools categorized in bars or panels → Build website
  4. Uploading – Upload any required files → Web server through Files Panel

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